Nikolaos (Nikos) Zirogiannis

Paul H. O'Neill School of Public and Environmental Affairs, Indiana University Bloomington
1315 East Tenth Street, Room 331, Bloomington, IN 47405
Office: (812) 856-8323; e-mail: nzirogia@iu.edu; web-page
-Updated July 2026-

Professional Appointments

Associate Professor, Paul H. O'Neill School of Public and Environmental Affairs, Indiana University Bloomington, 2026 to present
Assistant Professor, Paul H. O'Neill School of Public and Environmental Affairs, Indiana University Bloomington, 2020 to 2026
Affiliated Faculty, Integrated Program in the Environment, Indiana University Bloomington, 2015-2026
Assistant Scientist, Paul H. O'Neill School of Public and Environmental Affairs, Indiana University Bloomington, 2015 to 2020
Visiting Assistant Professor, School of Public and Environmental Affairs, Indiana University Bloomington, 2013 to 2015

Education

Ph.D. Resource Economics, University of Massachusetts Amherst, 2013
M.S. Resource Economics, University of Massachusetts Amherst, 2008
M.Sc. Environmental Sustainability, University of Edinburgh, U.K., 2006
B.A. Business Administration, Athens University of Economics and Business, Greece, 2003

Fields of Interest

Applied Econometrics, Environmental and Natural Resource Economics, Environmental Policy, Energy Policy, Water Policy.

Research

Peer Reviewed Papers

(Asterisks indicate graduate student co-authors)

21. **Zirogiannis, N.**, *Byrne, A., Hollingsworth, A., and Konisky, D.M. (2023). Polluting Under the Radar: Emissions, Inequality, and Concrete Batch Plants in Houston. *Environmental Science and Technology* 57, 11410-11419 https://doi.org/10.1021/acs.est.3c04412.
20. Hollingsworth, A., Konisky, D.M., and **Zirogiannis, N.** (2023). Excess Emissions: Environmental Impacts, Health Effects, and Policy Debate. *Review of Environmental Economics and Policy* 17(1) https://doi.org/10.1086/723885.
19. Hollingsworth, A., Konisky, D.M., and **Zirogiannis, N.** (2021). The Health Consequences of Excess Emissions: Evidence from Texas. *Journal of Environmental Economics and Management* 108 https://doi.org/10.1016/j.jeem.2021.102449.
18. **Zirogiannis, N.,** Simon, D., and Hollingsworth, A. (2020). Estimating Co-Pollutant Benefits from Climate Change Policies in the Electricity Sector: An Empirical Approach. *Energy Economics* 90 https://doi.org/10.1016/j.eneco.2020.104863.
17. Stranlund, J.K., Murphy, J.J., Spraggon, J.M., and **Zirogiannis, N.** (2019) Tying Enforcement to Prices in Emissions Markets: An Experimental Evaluation. *Journal of*

Environmental Economics and Management 98 https://doi.org/10.1016/j.jeem.2019.07.001.

16. *Li, Z., Konisky, D.M., and **Zirogiannis, N.** (2019). Racial, Ethnic and Income Disparities in Air Pollution: A Study of Excess Emissions in Texas. *PLOS ONE* https://doi.org/10.1371/journal.pone.0220696
15. Carley, S., **Zirogiannis, N.**, Siddiki, S., Duncan, D., and Graham, J.D. (2019) Overcoming the Shortcomings of Plug-In Electric Vehicle Policies. *Renewable and Sustainable Energy Reviews* 113: 109291 https://doi.org/10.1016/j.rser.2019.109291
14. Carley, S., **Zirogiannis, N.**, Duncan, D., Siddiki, S., and Graham, J. D. (2019) The Macroeconomic Effects of the 2017-2025 Federal Fuel Economy and Greenhouse Gas Emissions Standards. *Journal of Policy Analysis and Management,* 38(3): 732-763 https://doi.org/10.1002/pam.22132
13. **Zirogiannis, N.**, Duncan, D., Carley, S., Siddiki, S., and Graham, J. D. (2019). The Effect of Corporate Average Fuel Economy Standards on Vehicle Sales Projections: A Total Cost of Ownership Approach. *Transport Policy,* 75: 70-87 https://doi.org/10.1016/j.tranpol.2019.01.006
12. **Zirogiannis, N.**, Krutilla, K., Tripodis, Y., and *Fledderman, K. (2019). Human Development Over Time: An Empirical Comparison of a Dynamic Index and the Standard HDI. *Social Indicators Research*, 142(2): 773-798 https://doi.org/10.1007/s11205-018-1926-z
11. Duncan, D., *Ku, A., *Julian, A., Carley, S., Siddiki, S., **Zirogiannis, N.**, and Graham, J.D. (2019). Most Consumers Don't Buy Hybrids: Is Rational Choice a Sufficient Explanation? *Journal of Benefit-Cost Analysis,* 10(1): 1-38 https://doi.org/10.1017/bca.2018.24
10. Jenn, A., Hardman, S., Carley, S., **Zirogiannis, N.**, Duncan, D., and Graham, J.D. (2019). Cost Implications for Automaker Compliance with Zero Emissions Vehicle Requirements. *Environmental Science and Technology,* 53(2): 564-574 https://doi.org/10.1021/acs.est.8b03635
9. Carley, S., Davies, L., Spence, D., and **Zirogiannis, N.** (2018). Empirical Evaluation of the Stringency and Design of Renewable Portfolio Standards. *Nature Energy* 3: 754-763 https://doi.org/10.1038/s41560-018-0202-4
8. Siddiki, S., Carley, S., **Zirogiannis, N.**, Duncan, D., and Graham, J. D. (2018). Does Dynamic Federalism Yield Compatible Policies? A Study of the Designs of Federal and State Vehicle Policies. *Policy Design and Practice* 1(3): 215-232 https://doi.org/10.1080/25741292.2018.1505186
7. **Zirogiannis, N.,** Hollingsworth, A., and Konisky, D.M. (2018). Understanding Excess Emissions from Industrial Facilities: Evidence from Texas. *Environmental Science and Technology* 52(5): 2482-2490 https://doi.org/10.1021/acs.est.7b04887
6. **Zirogiannis, N.,** and Tripodis, Y. (2018). Dynamic Factor Analysis for Short Panels: Estimating Performance Trajectories for Water Utilities. *Statistical Methods and Applications* 27(1): 131-150 https://doi.org/10.1007/s10260-017-0394-y
5. Tripodis, Y., Alosco, M.L., **Zirogiannis, N.**, Gavett, B.E., Chaisson, C., Martin, B., McClean, M.D., Mez, J., Kowall, N., and Stern, R.A. (2017). The Effect of Traumatic Brain Injury History with Loss of Consciousness on Rate of Cognitive Decline Among Older Adults with Normal Cognition and Alzheimer's Disease Dementia. *Journal of Alzheimer's Disease* 59 (1): 251-263 https://doi.org/10.3233/JAD-160585

4. Timmons, D., **Zirogiannis, N.** and *Lutz, M. (2016). Location Matters: Population Density and Carbon Emissions from U.S. Residential Building Energy Use, *Energy Research and Social Science* 22: 137-146 https://doi.org/10.1016/j.erss.2016.08.011
3. **Zirogiannis, N.**, *Alcorn, J., Rupp, J., Carley, S., and Graham, J.D. (2016) State Regulations of Unconventional Gas Development in the U.S.: An Empirical Evaluation. *Energy Research and Social Science* 11: 142-154 https://doi.org/10.1016/j.erss.2015.09.009
2. **Zirogiannis, N.,** *Alcorn, J., *Piepenburg, J. and Rupp, J. (2015) I Want in on That: Community-Level Policies for Unconventional Gas Development in New York. *Agricultural and Resource Economics Review* 44 (2): 164-194 https://doi.org/10.1017/S1068280500010273
1. Tripodis, Y. and **Zirogiannis, N.** (2015) Dynamic Factor Analysis for Multivariate Time Series: An Application to Cognitive Trajectories. *International Journal of Clinical Biostatistics and Biometrics* 1 (1): 1-8 https://doi.org/10.23937/2469-5831/1510001

Book Chapters

Konisky, D.M., **Zirogiannis, N.**, and Carley, S. (2022). Building Resilience to the Energy Transition in Indiana, in *Climate Change and Resilience: Lessons for Hoosiers and Beyond*, McCabe, J.G., Filippelli, G.M., Novick, K.A., and Shanahan, J., eds. Indiana University Press.

Working papers

Hollingsworth, A., Ortega, A. and **Zirogiannis, N.** (2026). Decoding Success: Evidence from a Federal Phonics Initiative.

Hollingsworth, A., Ortega, A. and **Zirogiannis, N.** (2026). Educational Impacts of Air Pollution Exposure.

Work in progress

*Choi, J., Ortega, A. and **Zirogiannis, N.** (2025). Blades of Despair? The Impact of Wind Farms on Crime.

Peer Reviewed Policy Reports

Carley, S., Duncan, D., Graham, J. D., Siddiki, S., and **Zirogiannis, N.**, 2017. "A Macroeconomic Study of Federal and State Automotive Regulations: With Recommendations for Analysts, Regulators, and Legislators." [Peer-reviewed document]. Published by SPEA.

Carley, S., Duncan, D., Esposito, D., Graham, J. D., Siddiki, S., and **Zirogiannis, N.**, 2016. "Rethinking Auto Fuel Economy: Technical and Policy Suggestions for the 2016-17 Midterm Reviews." [Peer-reviewed document]. Published by SPEA.

Op-Eds, Public Comments

Hollingsworth, A., Konisky, D.M., and **Zirogiannis, N.** (2023). Public comments to the US Environmental Protection Agency on the Revisions to the Air Emissions Reporting Requirements (Docket ID No EPA–HQ-OAR–2004–0489) link

Zirogiannis, N., Hollingsworth, A., and Konisky, D.M. (2023). Comments to the Texas Commission on Environmental Quality on the 2023 Amendment to the Concrete Batch Plant Standard Permit (Non-Rule Project No. 2022-033-OTH-NR) link

Zirogiannis, N. (2022). "Regulatory Agencies can Improve Excess Emissions Tracking to Protect Health" Key Findings Brief, Scholars Strategy Network link

Hollingsworth, A., Konisky, D.M., and **Zirogiannis, N.** (2019). Public comments submitted to the US Environmental Protection Agency on the Withdrawal of Finding of Substantial Inadequacy of Implementation Plan and of Call for Texas State Implementation Plan Revision-Affirmative Defense Provisions (Docket ID No EPA–R06–OAR–2018–0770) link

Zirogiannis, N., Hollingsworth, A., and Wing, C. (2018) "Where did all the health benefits go? Evaluating EPA's repeal of the Clean Power Plan", SPEA Advance link.

Zirogiannis, N., Hollingsworth, A., and Konisky, D.M. (2018). "Air pollution from industrial shutdowns and startups worse than thought", The Conversation link
Reprinted in the Chicago Tribune (2/14/18), and the Houston Chronicle (2/14/18)

Grants

"Clinical and Social Determinants of Health in Extreme Heat" Co-PI with Sonia Angell, Dana Habeeb, Alex Hollingsworth, David Konisky, Kosali Simon and Coady Wing (PI). Indiana University Research through the Environmental Research Network. $30,000. 2024-2025

"The Educational Consequences of Excess Emissions" Co-PI with Alberto Ortega. Spencer Foundation Small Grants Program. $49,500. 2023-2028

"Community Air Quality Networks and Health Outcomes: A comparison of two study sites" PI with Co-PIs Alex Hollingsworth, David Konisky and Gabriel Filippelli. Indiana University Faculty Research Support Program-Seed Funding. $26,670. 2021-2022

"Study of the macro-economic impact of the light-duty vehicle corporate average fuel economy, greenhouse gas and zero-emission vehicle standards: Phases II and III" Co-PI with John Graham (PI), Sanya Carley, Denvil Duncan and Saba Siddiki. Alliance of Automobile Manufacturers. $590,000. 2016-2017

Invited Talks, Webinars, Panel Presentations

Excess Emissions: Environmental Impacts, Health Effects, and Policy Debate.
Webinar presentation at the Environmental Protection Agency, National Center for Environmental Economics, November 2nd 2023.
Department of Resource Economics, University of Massachusetts, Amherst, September 29th 2023.

The Environmental and Health Impacts of Excess Emissions: Evidence from Texas. Webinar presentation at the Environmental Protection Agency, Office of Air and Radiation, Aug. 19th 2021

State Regulations of Unconventional Gas Development in the U.S.: An Empirical Evaluation. Invited talk at the IU Geography Department, December 4th 2015.

Selected Conference Presentations

Hollingsworth, A., Ortega, A. and **Zirogiannis, N.** The Power of Reading: How Classroom Pedagogy and Out-of-Class Environment Combine to Affect Short-Run Learning and Later-in-Life Outcomes
2025: Association of Environmental and Resource Economists Annual Summer Conference, May 28-30, Santa Ana Pueblo, New Mexico

2024: Heartland Environmental and Resource Economic Workshop at the University of Illinois, Urbana-Champaign (Oct. 26-27)
2024: Northeastern Agricultural and Resource Economics Association Annual Conference, June 9-11 2024, Rehoboth Beach, Delaware

Zirogiannis, N., *Byrne, A., Hollingsworth, A., and Konisky, D.M. Emitting Under the Radar: The Environmental Justice Implications of Concrete Batch Plants in Houston,
2023: Northeastern Agricultural and Resource Economics Association Conference, June 11-13, Annapolis, Maryland
2022: Association for Public Policy Analysis and Management Fall Conference, November 17-19, Washington D.C.
2022: Association of Environmental and Resource Economists Annual Summer Conference, June 1-3, Miami, Florida

Hollingsworth, A., Konisky, D.M., and **Zirogiannis, N.** The Health Consequences of Excess Emissions: Evidence from Texas,
2021: 85th Annual Midwest Economics Association Conference (Virtual), March 24th; AAEA & WAEA Joint Annual Meeting (Hybrid) August 1-3 Austin, Texas
2020: Midwest Applied Microeconomics Workshop, Purdue University, February 28, West Lafayette, IN
2019 (poster): Association of Environmental and Resource Economists Annual Summer Conference, May 30-31, Incline Village, NV

Zirogiannis, N., Duncan, D., Carley, S., Siddiki, S., and Graham, J. D. (2017). The Effect of Corporate Average Fuel Economy Standards on Vehicle Sales Projections: A Total Cost of Ownership Approach. Association of Environmental and Resource Economists Annual Meeting May 31-June 2, Pittsburgh, PA

Zirogiannis, N., Hollingsworth, A., and Konisky, D.M. (2017). An Empirical Analysis of Excess Emissions from Industrial Facilities, Northeastern Agricultural and Resource Economics Association meeting June 11-13 2017, Washington, DC

Simon, D. and **Zirogiannis, N.** (2015). Greenhouse Gas Emissions and Criteria Air Co-Pollutants: Evidence from US Power Plants. Northeastern Agricultural and Resource Economics Association meeting June 28-30 2015, Newport, RI

Publicity and Media Mentions

Strott, S. (2023) Fine Particulates Kill Thousands of Texans a Year. It's Likely to Get Worse, Public Health Watch (link)

Resources Radio (2023). "Consequences of Unplanned and Under-Measured Air Pollution, with Nikos Zirogiannis", Podcast Interview (link)

Martinez, A. (2023). "Senator's Bill Would Fine Texans for Multiple Environmental Complaints that don't lead to enforcement", The Texas Tribune (link)

Bikoff, K. (2023). "O'Neill Researchers Urge EPA to Review Regulations Surrounding Excess Emissions", O'Neill School blog (link)

Strott, S. (2022). "There are 'No Real Consequences' for Toxic Emissions Events" Texas Observer (link)

Levy, M. G. (2021). "How Many People Die When Polluters Exceed their Limits?" WIRED (link)

IU Newsroom (2018). "States Boost Renewable Energy and Economic Development When Utilities Adopt Renewable Standards", July 23rd (link)

Fogel, B. (2018), "News Roundup: Researchers Estimate Impact of Excess emissions on Texans' Health", Texas Standard, Feb. 16th (link)

IU Newsroom (2018). "Excess emissions make significant contribution to air pollution, study finds", Feb. 14th (link)

Reilly, S. (2018). All states must track "excess emissions" – study, E&E News, Feb. 15th (link)

Teaching

- P710 (co-taught): **PhD Seminar in Environmental Policy**, Spring 2023, Spring 2026
- R674: **Energy Economics and Policy**, Spring 2021-2026
- R532: **Water Policy and Economics**, Fall 2014, Spring 2016, Fall 2018-2025
- E400/R535: **International Environmental Policy**, Spring 2014-2015, Fall 2020-2025
- E183: **Environment and People,** Spring 2014, Fall 2014, Spring 2019-2026
- INTL-I 300 (Honors): **International Energy Markets: Environmental, Economic and Health Aspects**, Fall 2018-2019
- INTL-I 300 (Honors): **Global Environmental Institutions and Development**, Fall 2016-2017

Professional Service

Referee and Reviewer Service

- American Journal of Health Economics, Ecological Economics, Economic Inquiry, Energy Economics, Energy Journal, Energy Policy, Energy Research and Social Science, Environmental Pollution, Environmental and Resource Economics, Journal of the Association of Environmental and Resource Economists, Journal of Environmental Economics and Management, Journal of Policy Analysis and Management, Land Economics, Public Budgeting and Finance, USDA Agriculture and Food Research Initiative Grant Review Panel Member (2023)

School and University Service

Member, Executive Committee of the Integrated Program in the Environment, IU Bloomington, 2021-2026

Coordinator of O'Neill School Environmental Policy Seminar Series, 2020-present

Member, Dual Degree Curriculum Committee, O'Neill SPEA, 2016-present

Member, PhD qualifying exam committee in Environmental Policy, O'Neill SPEA, 2013-2015, 2020-present

Member, PhD in Public Affairs admission committee, O'Neill SPEA, 2024-present

SPEA 183 Environment and People lead instructor, O'Neill SPEA, 2020-2024

Member, Master's in Public Affairs Admissions Committee, O'Neill SPEA, 2019

Advisory Board Member, Campus Writing Program, IU Bloomington, 2016-2020

Professional Development

Northeastern Agricultural and Resource Economics Association Career Mentoring Program, 2023-2025 cohort (mentor)

Scholars Strategy Network & Robert Wood Johnson Foundation Health Equity Scholars Program 2021-2022 cohort (mentee)

Northeastern Agricultural and Resource Economics Association Career Mentoring Program, 2015-2017 cohort (mentee)
Course Development Institute, Center for Innovative Teaching and Learning, Indiana University Bloomington, June 2014

Honors and Awards

Trustees Teaching Award, O'Neill SPEA, IU Bloomington, 2022
Graduate Teaching Award, O'Neill SPEA, IU Bloomington, 2021

Consulting

Short-term consultant for the World Bank, Washington, DC, August 2010-March 2011

Professional Society Memberships and Affiliations

Association for Public Policy Analysis and Management
Association of Environmental and Resource Economists
Northeastern Agricultural and Resource Economics Association